

BILKATE 5 mg Tablet

BETA-BLOCKING AGENT, SELECTIVE

Each tablet contains:
Nebivolol hydrochloride eq. to
Nebivolol 5 mg

PRODUCT DESCRIPTION:

White, round, biconvex tablet with engraved NB on one side and plain on the other
The tablet is not meant to be split and should be swallowed whole.

MECHANISM OF ACTION:

Nebivolol is a racemate of two enantiomers, SRRR-nebivolol (d-nebivolol) and RSSS-nebivolol (l-nebivolol). It combines two pharmacological activities:

- It is a competitive and selective beta-receptor antagonist: this effect is attributed to the SRRR-enantiomer (d-enantiomer).
- It has mild vasodilating properties, due to an interaction with the L-arginine/Nitric oxide pathway.

Single and repeated doses of Nebivolol reduce heart rate and blood pressure at rest and during exercise, both in normotensive subjects and in hypertensive patients. The antihypertensive effect is maintained during chronic treatment.

At therapeutic doses, Nebivolol is devoid of alpha-adrenergic antagonism.

During acute and chronic treatment with Nebivolol in hypertensive patients, systemic vascular resistance is decreased. Despite heart rate reduction, reduction in cardiac output during rest and exercise may be limited due to an increase in stroke volume. The clinical relevance of these hemodynamic differences as compared to other beta-1 receptor antagonists has not been fully established.

In hypertensive patients, Nebivolol increases the NO-mediated vascular response to Acetylcholine (ACh) which is reduced in patients with endothelial dysfunction.

The effect of Nebivolol was independent from age, gender, or left ventricular ejection fraction of the population.

Nebivolol has no intrinsic sympathicomimetic activity and has no membrane stabilizing action.

Nebivolol has no significant effect on maximal exercise capacity or endurance.

Pharmacokinetics

Absorption

Both Nebivolol enantiomers are rapidly absorbed after oral administration. The absorption of Nebivolol is not affected by food. Nebivolol can be given with or without meals.

Biotransformation

Nebivolol is extensively metabolized, partly to active hydroxy-metabolites. Nebivolol is metabolized via alicyclic and aromatic hydroxylation, N-dealkylation and glucuronidation; in addition, glucuronides of the hydroxy-metabolites are formed. The metabolism of Nebivolol by aromatic hydroxylation is subject to the CYP2D6 dependent genetic oxidative polymorphism. The oral bioavailability of Nebivolol averages 12% in fast metabolizers and is virtually complete in slow metabolizers. At steady state and at the same dose level, the peak plasma concentration of unchanged Nebivolol is about 23 times higher in poor metabolizers than in extensive metabolizers. When unchanged drug plus active metabolites are considered, the difference in peak plasma concentrations is 1.3 to 1.4 fold. Because of the variation in rates of metabolism, the dose of Nebivolol should always be adjusted to the individual requirements of the patient: Poor metabolizers therefore may require lower doses. In fast metabolizers, elimination half-lives of the Nebivolol enantiomers average 10 hours. In slow metabolizers, they are 3-5 times longer. In fast metabolizers, plasma levels of the RSSS-enantiomer are slightly higher than for the SRRR-enantiomer. In slow metabolizers, this difference is larger. In fast metabolizers, elimination half-lives of the hydroxymetabolites of both enantiomers average 24 hours, and are about twice as long in slow metabolizers. Steady state plasma levels in most subjects (fast metabolizers) are reached within 24 hours for Nebivolol and within a few days for the hydroxy-metabolites. Plasma concentrations are dose-proportional between 1 and 30 mg. The pharmacokinetics of Nebivolol are not affected by age.

Distribution

In plasma, both Nebivolol enantiomers are predominantly bound to albumin. Plasma protein binding is 98.1% for SRRR-nebivolol and 97.9% for RSSS-nebivolol.

Elimination

One week after administration, 38% of the dose is excreted in the urine and 48% in the feces. Urinary excretion of unchanged Nebivolol is less than 0.5% of the dose.

INDICATION:

Hypertension

Treatment of essential hypertension.

Chronic heart failure (CHF)

Treatment of stable mild and moderate chronic heart failure in addition to standard therapies in elderly patients ≥ 70 years.

DOSAGE AND ADMINISTRATION:

Oral

Posology

Hypertension

Adults

The dose is one tablet (5 mg) daily, preferably at the same time of the day.

The blood pressure lowering effect becomes evident after 1-2 weeks of treatment. Occasionally, the optimal effect is reached only after 4 weeks.

Combination with other antihypertensive agents

Beta-blockers can be used alone or concomitantly with other antihypertensive agents. Till date, an additional antihypertensive effect has been observed only when Nebivolol 5 mg is combined with Hydrochlorothiazide 12.5-25 mg.

Patients with renal insufficiency

In patients with renal insufficiency, the recommended starting dose is 2.5 mg daily. If needed, the daily dose may be increased to 5 mg.

Patients with hepatic insufficiency

Data in patients with hepatic insufficiency or impaired liver function are limited. Therefore, the use of Nebivolol in these patients is contraindicated.

Older people

In patients over 65 years, the recommended starting dose is 2.5 mg daily. If needed, the daily dose may be increased to 5 mg. However, in view of the limited experience in patients above 75 years, caution must be exercised and these patients monitored closely.

Pediatric population

The efficacy and safety of Nebivolol in children and adolescents aged below 18 years has not been established. No data are available. Therefore, use in children and adolescents is not recommended.

Chronic heart failure (CHF)

The treatment of stable chronic heart failure has to be initiated with a gradual up-titration of dosage until the optimal individual maintenance dose is reached. Patients should have stable chronic heart failure without acute failure during the past six weeks. It is recommended that the treating physician should be experienced in the management of chronic heart failure. For those patients receiving cardiovascular drug therapy including diuretics and/or Digoxin and/or ACE inhibitors and/or angiotensin II antagonists, dosing of these drugs should be stabilized during the past two weeks prior to initiation of Nebivolol treatment.

The initial up-titration should be done according to the following steps at 1-2 weekly intervals based on patient tolerability: 1.25 mg Nebivolol, to be increased to 2.5 mg Nebivolol once daily, then to 5 mg once daily and then to 10 mg once daily. The maximum recommended dose is 10 mg Nebivolol once daily.

Initiation of therapy and every dose increase should be done under the supervision of an experienced physician over a period of at least 2 hours to ensure that the clinical status (especially as regards blood pressure, heart rate, conduction disturbances, signs of worsening of heart failure) remains stable. Occurrence of adverse events may prevent all patients being treated with the maximum recommended dose. If necessary, the dose reached can also be decreased step by step and reintroduced as appropriate. During the titration phase, in case of worsening of the heart failure or intolerance, it is recommended first to reduce the dose of Nebivolol, or to stop it immediately if necessary (in case of severe hypotension, worsening of heart failure with acute pulmonary edema, cardiogenic shock, symptomatic bradycardia or AV block). Treatment of stable chronic heart failure with Nebivolol is generally a long-term treatment. The treatment with Nebivolol is not recommended to be stopped abruptly since this might lead to a transitory worsening of heart failure. If discontinuation is necessary, the dose should be gradually decreased divided into halves weekly.

Patients with renal insufficiency

No dose adjustment is required in mild to moderate renal insufficiency since up-titration to the maximum tolerated dose is individually adjusted. There is no experience in patients with severe renal insufficiency (serum creatinine ≥ 250 $\mu\text{mol/L}$). Therefore, the use of Nebivolol in these patients is not recommended.

Patients with hepatic insufficiency

Data in patients with hepatic insufficiency are limited. Therefore the use of Nebivolol in these patients is contraindicated.

Older people

No dose adjustment is required since up-titration to the maximum tolerated dose is individually adjusted.

Pediatric population

The efficacy and safety of Nebivolol in children and adolescents aged below 18 years has not been established. No data are available. Therefore, use in children and adolescents is not recommended.

Method of administration

Tablets may be taken with meals.

PRECAUTION:

The following warnings and precautions apply to beta-adrenergic antagonists in general:

Anesthesia

Continuation of beta blockade reduces the risk of arrhythmias during induction and intubation. If beta blockade is interrupted in preparation for surgery, the beta-adrenergic antagonist should be discontinued at least 24 hours beforehand. Caution should be observed with certain anesthetics that cause myocardial depression. The patient can be protected against vagal reactions by intravenous administration of Atropine.

Cardiovascular

In general, beta-adrenergic antagonists should not be used in patients with untreated congestive heart failure (CHF), unless their condition has been stabilized. In patients with ischemic heart disease, treatment with a beta-adrenergic antagonist should be discontinued gradually, i.e. over 1-2 weeks. If necessary replacement therapy should be initiated at the same time, to prevent exacerbation of angina pectoris. Beta-adrenergic antagonists may induce bradycardia: if the pulse rate drops below 50-55 bpm at rest and/or the patient experiences symptoms that are suggestive of bradycardia, the dosage should be reduced.

Beta-adrenergic antagonists should be used with caution:

- in patients with peripheral circulatory disorders (Raynaud's disease or syndrome, intermittent claudication), as aggravation of these disorders may occur;
- in patients with first degree heart block, because of the negative effect of beta-blockers on conduction time;
- in patients with Prinzmetal's angina due to unopposed alpha receptor-mediated coronary artery vasoconstriction: beta-adrenergic antagonists may increase the number and duration of anginal attacks.

Combination of Nebivolol with calcium channel antagonists of the Verapamil and Diltiazem type, with Class I antiarrhythmic drugs, and with centrally acting antihypertensive drugs is generally not recommended (see Drug interaction).

Metabolic/ Endocrinological

Nebivolol does not affect glucose levels in diabetic patients. Care should be taken in diabetic patients, however, as Nebivolol may mask certain symptoms of hypoglycemia (for example tachycardia and palpitations).

Beta-adrenergic blocking agents may mask tachycardic symptoms in hyperthyroidism. Abrupt withdrawal may intensify symptoms.

24 cm.
Front
17 cm.

Product	Code No.	Dimension	Packaging Type	Thickness
BILKATE 5	IBMY 0060	W 17 x L 24 cm.	Wood Free Paper (0.25 mm)	60 g (0.08 mm.)
PM Specification				
Designed by: (PDM, APD, RDCM, RDC-A, RDCM, RDC-A)	Checked by: (PDM, APD)		Approved by: (M.D. Only (Intern and Comm))	CUSTOMER/SALES DEPT: (Only)
Approved by: PLC (Only)	IRA: (Only)	GCC-PM: (Only)		
IRA: (Only)				

BILKATE 5_Insert_Front Back_IBMY 0060 BY: NUY 18/07/2023

